Chapter 4
Linear Transformations

4.1 Definition and Examples

Recall: vector spaces

- Given a vector space V, we can:
  - Add vectors: $\mathbf{v}_1 + \mathbf{v}_2 \in V$
  - Multiply by scalars: $\alpha\mathbf{v} \in V$
  - Take *linear combinations*: $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots \in V$
- Given vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, we can ask:
  - When will $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \ \ldots + \alpha_n\mathbf{v}_n$ be $\mathbf{0}$? (*Null space, linearly dependent/independent*)
  - What are all possible $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \ \ldots + \alpha_n\mathbf{v}_n$? (*Column space, span*)
  - Can all vectors $\mathbf{v} \in V$ be uniquely written as $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \ \ldots + \alpha_n\mathbf{v}_n$? (*Basis, coordinate vectors*)
- In this chapter, we will study "linear" functions between "linear" spaces.

Linear Transformation线性变换

Definition

A mapping L from a vector space V into a vector space W is said to be a ***linear transformation线性变换*** if

$$L(\alpha\mathbf{v}_1 + \beta\mathbf{v}_2) = \alpha L(\mathbf{v}_1) + \beta L(\mathbf{v}_2) \quad (1)$$

for all $\mathbf{v}_1, \mathbf{v}_2 \in V$ and for all scalars α and β.

Alternative definition

L is a linear transformation if and only if for all $\mathbf{v}_1, \mathbf{v}_2 \in V$ and scalars α,

$$L(\mathbf{v}_1 + \mathbf{v}_2) = L(\mathbf{v}_1) + L(\mathbf{v}_2) \quad (2)$$

and

$$L(\alpha\mathbf{v}) = \alpha L(\mathbf{v}) \quad (3)$$

(1) ⇒(2)(3): Let α= β=1 for (2); Let β=0 for (3).

(2)(3)⇒(1): $L(\alpha\mathbf{v}_1 + \beta\mathbf{v}_2) = L(\alpha\mathbf{v}_1) + L(\beta\mathbf{v}_2)$ [using (2)]

$= \alpha L(\mathbf{v}_1) + \beta L(\mathbf{v}_2)$ [using (3)]

Notation

- A mapping L from a vector space V into a vector space W will be denoted

$$L : V \rightarrow W$$

When the arrow notation is used, it will be assumed that V and V represent vector spaces.

- When the vector spaces V and W are the same, we will refer to a linear transformation $L : V \rightarrow V$ as a linear operator on V. Thus, a ***linear operator线性算子*** is a linear transformation that maps a vector space V into itself.

$$Linear\ operator \subset linear\ transformation$$

$$L: V \rightarrow V \qquad\qquad L: V \rightarrow W$$

Example 1: dilation 伸缩

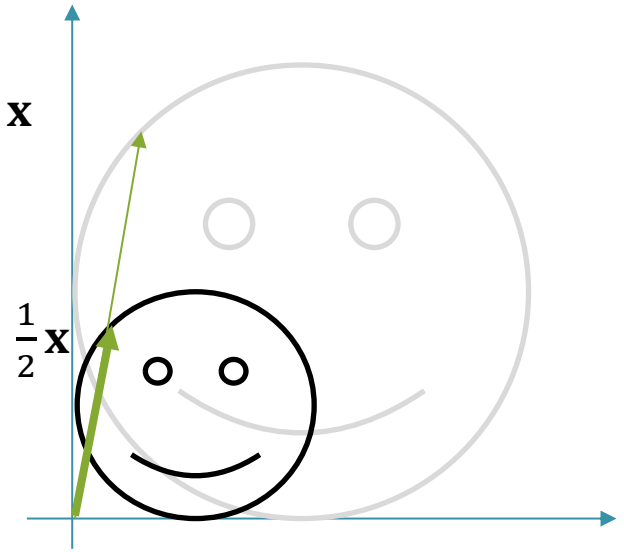
Show that $L: \mathbb{R}^2 \to \mathbb{R}^2$ via $L(\mathbf{x}) = \frac{1}{2}\mathbf{x}$ is a linear transformation.

Solution

For each $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$, and each α,

- $L(\alpha\mathbf{x}) = \frac{1}{2}(\alpha\mathbf{x}) = \alpha(\frac{1}{2}\mathbf{x}) = \alpha L(\mathbf{x})$
- $L(\mathbf{x}+\mathbf{y}) = \frac{1}{2}(\mathbf{x}+\mathbf{y}) = \frac{1}{2}\mathbf{x} + \frac{1}{2}\mathbf{y}$
  $= L(\mathbf{x}) + L(\mathbf{y})$

Therefore, $L(\mathbf{x})$ is a linear transformation.

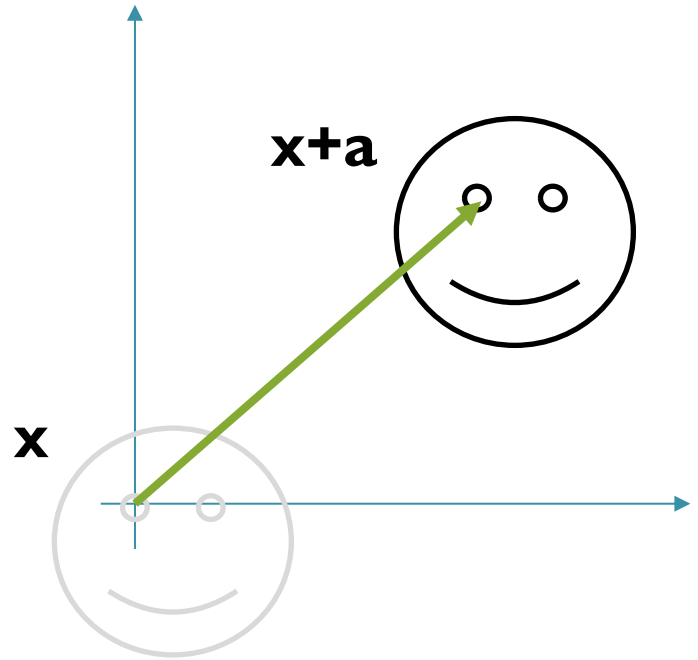


In general, if c is a positive scalar, the linear operator $F(\mathbf{x}) = c\mathbf{x}$ can be thought of as a stretching or shrinking by a factor of c.

Example 2: translation 平移

Let **a** be a fixed nonzero vector in R^2. A mapping of the form $L(\mathbf{x}) = \mathbf{x}+\mathbf{a}$ is called a translation. Show that a translation is **NOT** a linear transformation.

- Hint 1: calculate both $L(\mathbf{x}+\mathbf{y})$ and $L(\mathbf{x})+ L(\mathbf{y})$. Are they equal?
- Hint 2: calculate both $L(\alpha\mathbf{x})$ and $\alpha L(\mathbf{x})$. Are they equal?



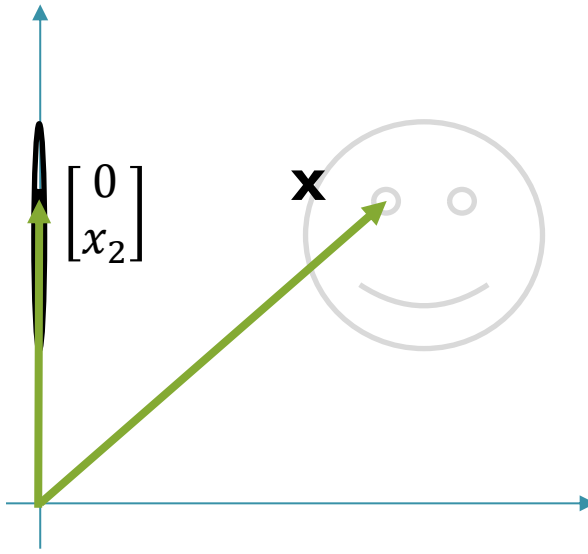
Example 3: projection 投影

Show that $L: \mathbb{R}^2 \to \mathbb{R}^2$ via $L\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ x_2 \end{bmatrix}$ is a linear operator. Describe L geometrically.

Solution

- For each $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2$, $L(\mathbf{x}) = \begin{bmatrix} 0 \\ x_2 \end{bmatrix}$, then

$$L(\alpha \mathrm{x}) = \begin{bmatrix} 0 \\ \alpha x_2 \end{bmatrix} = \alpha \begin{bmatrix} 0 \\ x_2 \end{bmatrix} = \alpha L(\mathrm{x})$$



- And for $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \in \mathbb{R}^2$,

$$L(\mathbf{x} + \mathbf{y}) = \begin{bmatrix} 0 \\ x_2 + y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ y_2 \end{bmatrix} = L(\mathbf{x}) + L(\mathbf{y})$$

- Then, it follows that $L(\mathbf{x})$ is a linear operator.

We can think of L as a projection onto the x_2- $axis$.

Example 4: rotation 旋转

Show that $L: \mathbb{R}^2 \to \mathbb{R}^2$ via $L\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$ is a linear operator.

Describe L geometrically.

Solution

- For each $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2, \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \in \mathbb{R}^2$, since

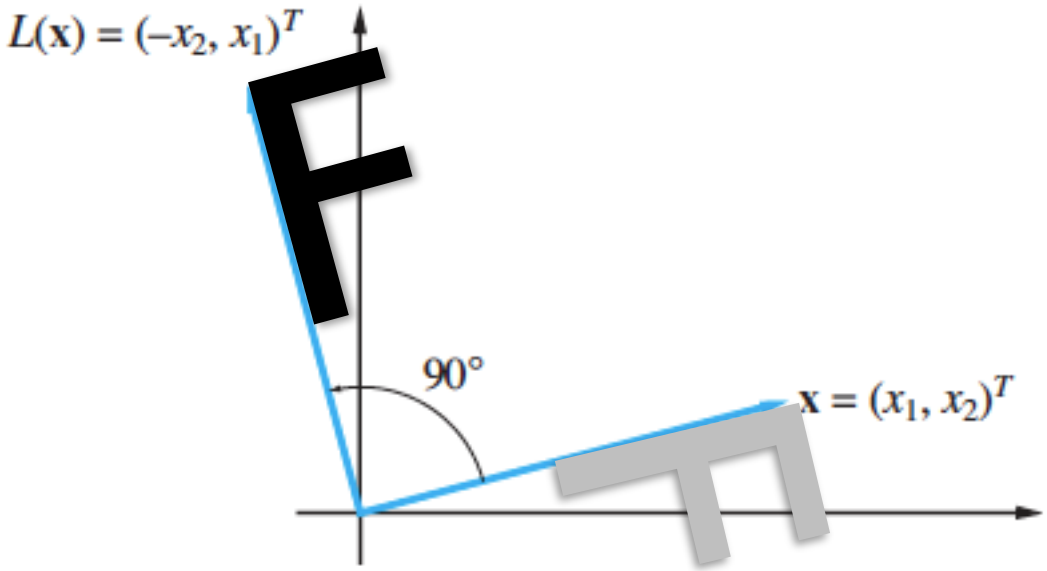
$$L(\alpha\mathbf{x} + \beta\boldsymbol{y}) = \left(\begin{bmatrix} -(\alpha x_2 + \beta y_2) \\ \alpha x_1 + \beta y_1 \end{bmatrix}\right) = \alpha \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix} + \beta \begin{bmatrix} -y_2 \\ y_1 \end{bmatrix}$$
$$= \alpha L(\mathbf{x}) + \beta L(\boldsymbol{y})$$

- Then, it follows that $L(\mathbf{x})$ is a linear operator.

see next page

Example 4: rotation 旋转

Show that $L: \mathbb{R}^2 \to \mathbb{R}^2$ via $L\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ is a linear operator. Describe L geometrically.



The operator of L has the effect of rotating each vector in $\mathbb{R}^2$ by 90° in the counterclockwise direction.

Example

$C^1[a, b]$: the space of functions on [a, b] whose derivative is continuous.

$C\ [a, b]$: the space of functions on [a, b] that is continuous.

Is $L: C^1[a, b] \to C\ [a, b]$ via $L(f) = f'$ a linear transformation?

Example

$C^1[a, b]$: the space of functions on [a, b] whose derivative is continuous.

$C\ [a, b]$: the space of functions on [a, b] that is continuous.

Is $L: C^1[a, b] \to C\ [a, b]$ via $L(f) = f'$ a linear transformation?

Solution

- $L(\alpha f + \beta g) = (\alpha f + \beta g)' = \alpha f' + \beta g' = \alpha L(f) + \beta L(g)$
- So, L is a linear transformation.

- What other "operations" in calculus are linear?
  → for example, what is $\lim_{x \to c}(\alpha f + \beta g)$?

Example

Is $L: \mathbb{R}^{n\times n} \to \mathbb{R}$ via $L(A) = \det(A)$ a linear transformation?

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Is $L: \mathbb{R}^{n\times n} \to \mathbb{R}$ via $L(A) = \det(A)$ a linear transformation?

Solution

- $L(\alpha A) = \det(\alpha A) = \alpha^n \det(A)$
- So, L is NOT a linear transformation.

Example

Is $L: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{n \times m}$ via $L(A) = A^T$ a linear transformation?

Example

Is $L: \mathbb{R}^{m\times n} \to \mathbb{R}^{n\times m}$ via $L(A) = A^T$ a linear transformation?

Solution

- Since

$$L(\alpha A + \beta B) = (\alpha A + \beta B)^T = \alpha A^T + \beta B^T$$
$$= \alpha L(A) + \beta L(B)$$

- Therefore, $L(A)$ is a linear transformation.

Followup: is $L(A) = A^{-1}$ a linear transformation?

Example

Let $C \in R^{n \times n}$ be a fixed matrix and define $L\colon R^{n \times n} \to R^{n \times n}$ via $L(A) = C^2 A$. Is L a linear operator?

Example

Let $C \in R^{n\times n}$ be a fixed matrix and define L: $R^{n\times n} \to R^{n\times n}$ via $L(A) = C^2A$. Is L a linear operator?

Solution

- Let A, B be any $n \times n$ matrix, and α be any scalar.
  - $L(A+B) = C^2(A+B) = C^2A + C^2B = L(A) + L(B)$
  - $L(\alpha A) = C^2(\alpha A) = \alpha C^2A = \alpha L(A)$
- Therefore, L is a linear operator.

Follow-up question: what about $L(A) = A^2C$?

Linear Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

If A is any $m \times n$ matrix, and define

$$L: \mathbb{R}^n \to \mathbb{R}^m$$

via

$$L(\mathbf{x}) = A\mathbf{x}$$

for all $\mathbf{x} \in \mathbb{R}^n$. Then L is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Thus, each $m \times n$ matrix corresponds in a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Question: does each linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ correspond to an $m \times n$ matrix? It is true: see section 4.2.

Linear Transformations from V to W

If L is a linear transformation mapping a vector space V into a vector space W, then

(1) $L(\mathbf{0}_V) = \mathbf{0}_W$ (where $\mathbf{0}_V$ and $\mathbf{0}_W$ are the zero vectors in V and W, respectively).

(2) If $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are elements of V and $\alpha_1, \ldots, \alpha_n$ are scalars, then $L(\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n) = \alpha_1 L(\mathbf{v}_1) + \alpha_2 L(\mathbf{v}_2) \ldots + \alpha_n L(\mathbf{v}_n)$

(3) $L(-\mathbf{v}) = -L(\mathbf{v})$ for all $\mathbf{v} \in V$.

Linear transformations send zero to zero, send linear combinations to linear combinations, and send opposite vectors to opposite vectors.

Also, if $L: V \to W$ has $L(\mathbf{0}_V) \neq \mathbf{0}_W$, then L is NOT a linear transformation. For example, this quickly shows that translation, $L(\mathbf{x}) = \mathbf{x} + \mathbf{a}$, is NOT a linear transformation.

Kernel核

Kernel Definition

Let $L: V \rightarrow W$ be a linear transformation. The ***kernel*** of L, denoted $\ker(L)$, is defined by

$$\ker(L) = \{\mathbf{v} \in V \mid L(\mathbf{v}) = \mathbf{0}_W\}$$

- The kernel of L is the set of all vectors from V that L transforms to the zero vector in W.
- Note, if L is defined via matrix multiplication for some matrix A, then $\ker(L) = N(A)$, the null space of A.

Image像 and Range

Image Definition

Let $L: V \rightarrow W$ be a linear transformation and let S be a subspace of V. The ***image*** of S, denoted $L(S)$, is defined by

$$L(S) = \{\mathbf{w} \in W \mid \mathbf{w} = L(\mathbf{v}) \text{ for some } \mathbf{v} \in S\}$$

The image of the entire vector space, $L(V)$, is called the ***range*** of L.

- If $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is a spanning set of S, then $L(S)$ is the span of $L(\mathbf{v}_1), L(\mathbf{v}_2), \ldots L(\mathbf{v}_k)$.

- If L is defined via matrix multiplication for some matrix A, then $L(V)$, the range of L, is the column space of A.

Kernel/Image are subspaces

Theorem

If $L: V \to W$ is a linear transformation and S is a subspace of V. Then

- $\ker(L) = \{\mathbf{v} \in V: L(\mathbf{v}) = \mathbf{0}_W\}$ is a subspace of V.
- $L(S) = \{\mathbf{w} \in W: \mathbf{w} = L(\mathbf{v}) \text{ for some } \mathbf{v} \in S\}$ is a subspace of W.

Proof

- We will prove $\ker(L)$ is a subspace of L.
- First, $\ker(L) \neq \emptyset$ since $\mathbf{0}_V \in$ker(L).
- Then for $\forall\, \mathbf{v_1}, \mathbf{v_2} \in \ker(L)$, that is $L(\mathbf{v_1}) = \mathbf{0}_W, L(\mathbf{v_2}) = \mathbf{0}_W$, then
- $L(\alpha\mathbf{v}_1 + \beta\mathbf{v}_2) = \alpha L(\mathbf{v}_1) + \beta L(\mathbf{v}_2) = \alpha\mathbf{0}_w + \beta\mathbf{0}_w = \mathbf{0}_w$, therefore $\alpha\mathbf{v}_1 + \beta\mathbf{v}_2 \in \ker(L)$. Therefore $\ker(L)$ is closed under scaler multiplication and addition.

Similarly, you can prove that $L(S)$ is a subspace of W.

Example

It can be shown that $L: \mathbb{R}^2 \to \mathbb{R}^2$ via $L\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x + y \\ 3x + y \end{bmatrix}$ is a linear operator. Find $\ker(L)$ and the range of L.

Example

It can be shown that $L: \mathbb{R}^2 \to \mathbb{R}^2$ via $L\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x + y \\ 3x + y \end{bmatrix}$ is a linear operator. Find $\ker(L)$ and the range of L.

Solution

- If we solve $L(\mathbf{x}) = 0$, we get:

$$2x + y = 0$$
$$3x + y = 0$$

- The only solution is $[\mathrm{x}, \mathrm{y}]^{\mathrm{T}} = [0,0]^T$, so $\ker(L)$={ **0** }.
- A vector $\mathbf{v} \in \mathbb{R}^2$ is in the range of L if and only if $\mathbf{v} = x\begin{bmatrix} 2 \\ 3 \end{bmatrix} + y\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Therefore $L(\mathbb{R}^2)$ is the space spanned by $[2,3]^T$ and $[1,1]^T$. As the two vectors are linearly independent, $L(\mathbb{R}^2) = \mathbb{R}^2$.

Example

It can be shown that $L: \mathbb{R}^3 \to \mathbb{R}^3$ via $L\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} x+y \\ y-z \\ x+z \end{bmatrix}$ is a linear operator.

$$L\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

1. Find the image of $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ -2 \\ -2 \end{bmatrix}$
2. Find $\ker(L)$ and the range of L.
3. Find $L(S)$, where $S = Span\left(\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}\right)$.
4. Find $L(S)$, where $S = Span\left(\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}\right)$.

Example

It can be shown that $L: \mathbb{R}^3 \to \mathbb{R}^3$ via $L\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} x+y \\ y-z \\ x+z \end{bmatrix}$ is a linear operator.

$$L\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

1. Find the image of $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ -2 \\ -2 \end{bmatrix}$

   1. $L\left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix}, L\left(\begin{bmatrix} 2 \\ -2 \\ -2 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

2. Find $\ker(L)$ and the range of L.

   2. Null space, column space (do it yourself)

3. Find $L(S)$, where $S = Span\left(\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}\right)$.

   3. Span of $L\left(\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

4. Find $L(S)$, where $S = Span\left(\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}\right)$.

   4. Span of $L\left(\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$ and $L\left(\begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$